
CE 523: Building Science

Course Code : CE 523

Course Name : Building Science

L-T-P-C : 3-0-0-3

Intended for : UG/PG

Prerequisite : Engineering Thermodynamics and Engineering Mathematics

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Building system** (6 Hours)

- Introduction: Life phases of a building; Resources utilized by buildings and the need for sustainability; Design goals and approach
- Outdoor environment: Air temperature and humidity, solar radiation, wind, precipitation and driving rain; Climatic zones, urban climate, site climate
- Indoor environment: Thermal comfort; Indoor air quality; Visual comfort; Effect of noise on wellness; Dangers associated with a building fire
- Building envelope: Walls, roofs, openings; Thermal mass, air and water tightness.

- **Moisture** (6 Hours)

- Psychrometrics: Moist air properties; Psychrometric chart
- Moisture transport mechanisms: Capillary action, permeation, vapor diffusion
- Moisture transport in building elements: Glaser method; Overview of transient models and their numerical solution Prevention of moisture uptake: Damp proofing, vapor barrier, role of thermal insulation and ventilation; Moisture meters .

- **Ventilation and Infiltration** (6 Hours)

- Natural ventilation: Aerodynamics around a building, ventilation due to wind and stack effects, cross-ventilation; Design guidelines for openings
- Infiltration: Crack flow equation; Calculation of leakages; Energy implications

- **Heat** (6 Hours)

- Mechanisms and laws of heat transfer: Conduction, Convection and Radiation
- Heat transfer in building elements: Steady state transfer in solid wall, wall with airspace, window, slab-on-grade, basement wall and floor, thermal bridges; Introduction to the continuous and distributed methods for analyzing transient heat flow
- Calculation of heating and cooling loads: Instantaneous, cooling load temperature difference/cooling load factor methods; Overview of transfer function, heat balance and radiant time series methods
- Thermal control: Overview of mechanical and structural controls

- **Acoustics** (6 Hours)

- Fundamentals: Wave nature of sound; Frequency, pressure, power and intensity of sound; Human perception of loudness; Environmental noise; Noise abatement; Sound meters
- Room acoustics: Direct and diffuse sound fields; Echo, standing wave; Types of sound absorbers; Sound absorption coefficient and Reverberation time
- Sound transmission and insulation: Resonance and Coincidence frequencies; Calculation of sound reduction index

- **Illumination** (6 Hours)

- Fundamentals: Human perception of light; Photometric quantities; Optical properties of matter; Light meters
- Daylight: Components of daylight; Solar altitude and azimuth, sun path diagram; Sky conditions and luminance; Illumination of horizontal and vertical surfaces; Daylight factor
- Artificial light: Electric light sources; Properties of light sources

- **Fire** (6 Hours)

- Fundamentals: Combustion process, development phases, fire classes, fire zones, smoke layer thickness, smoke temperature-time curve
- Fire safety: Response of building materials to fire and related performance classes; Fire resistance classification; Smoke control; General safety requirements

Textbooks:

1. Pinteric, M., **Building Physics: From physical principles to international standards**, 2nd edition, Springer, Switzerland, 2022.
2. Medved, S., **Building Physics: Heat, Ventilation, Moisture, Light, Sound, Fire, and Urban Microclimate**, Springer, 2022.

References:

1. Zhai, Z., **Energy Efficient Buildings: Fundamentals of Building Science and Thermal Systems**, Wiley, 2023.
2. Koenigsberger O.H., Ingersoll, T.G., Mayhew, A., and Szololay, S.V., **Manual of tropical housing and building**, Springer, 2022.
3. Moss, K.J., **Heat and Mass transfer in Buildings**, 2nd Edition, Taylor & Francis, 2007.
4. **ASHRAE, Handbook of Fundamentals**, American Society of Heating, Refrigerating and Air conditioning Engineers, 2021.
5. **BIS, National Building Code of India**, Bureau of Indian Standards, 2016.
6. **ECBC Code, Energy Conservation Building Code 2017**, Bureau of Energy Efficiency, 2017.